

3(a)(i)	<i>any two from:</i> (Neutrase® breaks down / hydrolyses) protein to, peptides / amino acids / smaller molecules ; A idea of increase in solubility during the reaction / AW, more light passes through / more light is transmitted / less light is absorbed ; <i>idea that 100 s is long enough to see the progress of the reactions ;</i> I 'the rate of reaction can be calculated' I 'allow time for reaction to complete'	2
3(a)(ii)	<i>accept ora where appropriate</i> 1 copper sulfate, decreases / AW, the activity of Neutrase (<i>ref. to A or B</i>) ; 2 0.01 (mol dm⁻³) / low concentration, CuSO₄ has less of an effect than, 0.05 (mol dm⁻³) / high concentration (<i>ref. to A and B</i>) ; 3 potassium sulfate has, little / no, effect on activity (<i>ref. to C</i>) ; 4 data quote to show absorbance for two different lines at the same time ; <i>one time and two absorbance readings from different lines on the graph with the unit for time used once anywhere in answer, allow 'at the end' for 100 s</i> 5 copper sulfate is an inhibitor of Neutrase ; 6 potassium sulfate is not an inhibitor ; A 'less of...' 7 copper sulfate binds to Neutrase ; A anywhere 8 substrate cannot enter active site / ESCs do not form (so protein not hydrolysed) ; A fewer ESCs	

Question	Answer	Marks
3(b)	<i>higher productivity / higher yield and fewer costs because</i> enzyme can be re-used ; enzyme can be easily recovered ; downstream processing is easier ; product, not / less, contaminated ; A less purification needed longer shelf-life of enzyme ; reduces product inhibition ; enzyme is, more stable / less likely to denature <i>or</i> described ; A thermostable / can work at high temperatures A in context of change in pH I 'can withstand changes in temperature'	2

Question	Answer	Marks
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